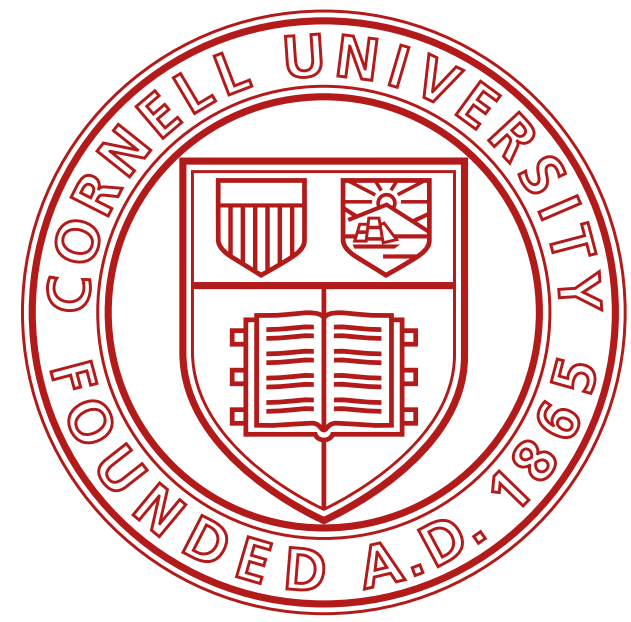


MoRE: Mixture of Reused Experts

Paper, code & contact
ericqiu6.github.io/MoRE



Eric S. Qiu^{*} · Utku Umur Acikalin^{*} · Justin Lovelace · Christian Belardi · Arjun B. Mulchandani · Carla P. Gomes · Kilian Q. Weinberger

Cornell University · Department of Computer Science · ^{*}Equal contribution

Experts don't scale across depth

Standard MoE gives every layer its **own private pool** of N experts. Each router picks K from its local pool, so the routing space is locked to $\binom{N}{K}$ combinations. Widening the pool means adding experts, which grows parameters linearly. Yet, layers with similar roles (e.g., early layers) relearn **redundant experts**. Hence, our intuition is that *some layer functionality should be reused, not relearned*.

1. Combine the experts from multiple layers into one pool, 2. then condition routing on depth

① Tie across layers

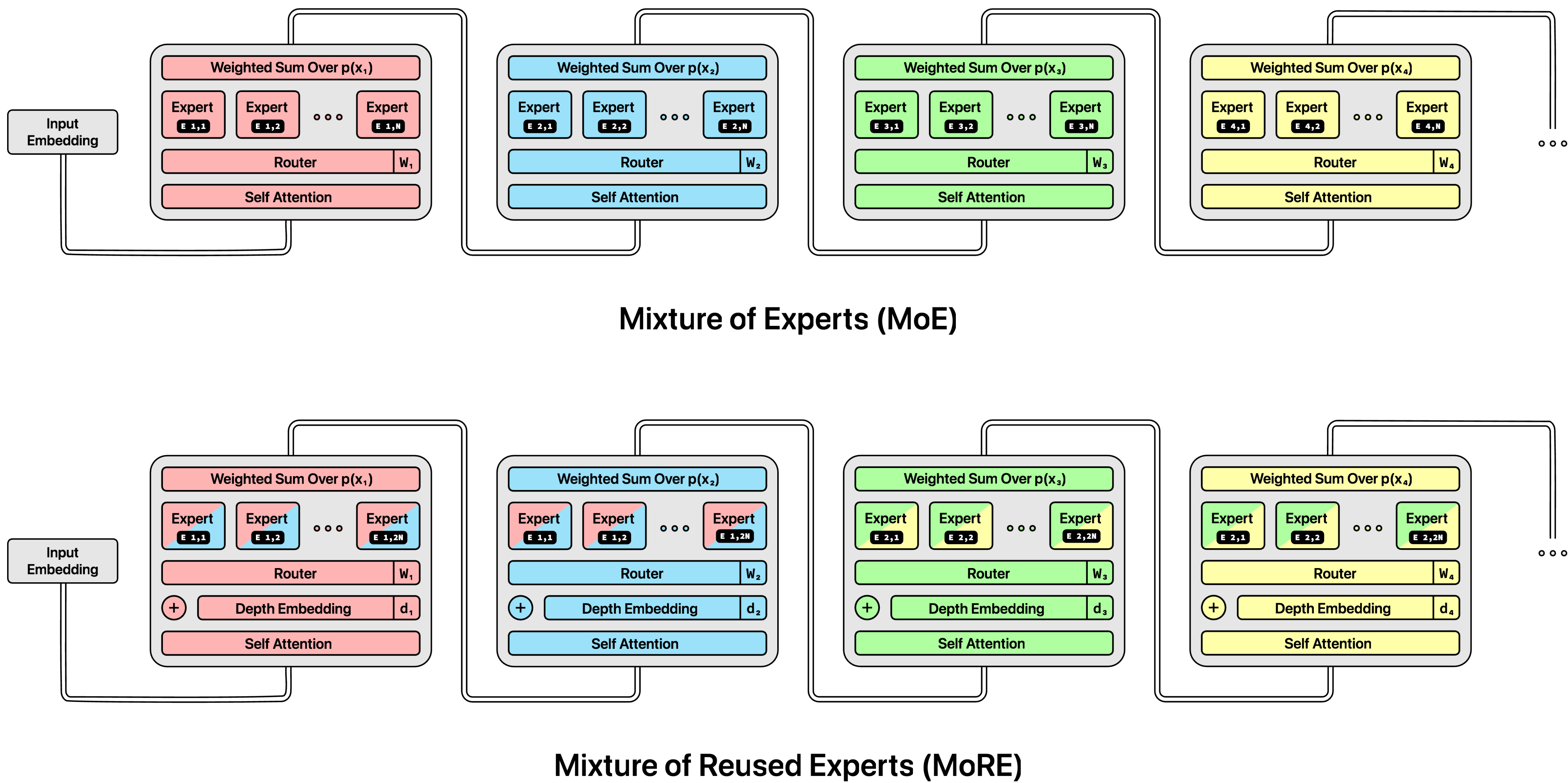
Adjacent layers (defined as a reuse block of R = 2 layers) draw from one shared pool of 2N experts. The total number of unique experts does not change (hence no additional parameter cost), but the routing space grows combinatorially.

$$\{E_1, \dots, E_N\} \rightarrow \{E_{b,1}, \dots, E_{b,R \cdot N}\}$$
$$\binom{N}{K} \rightarrow \binom{R \cdot N}{K} \text{ choices / layer}$$

② Depth embedding

Add a tiny learnable $\mathbf{d}_\ell$ to the input before routing. The same shared experts specialize to depth-conditioned input $\mathbf{x}_\ell + \mathbf{d}_\ell$, while the weights stay fully tied.

$$\mathbf{x}'_\ell = \mathbf{x}_\ell + \mathbf{d}_\ell$$
$$\mathbf{y}_\ell = \sum_{i \in T_\ell} p_i(\mathbf{x}'_\ell) E_{b,i}(\mathbf{x}'_\ell)$$

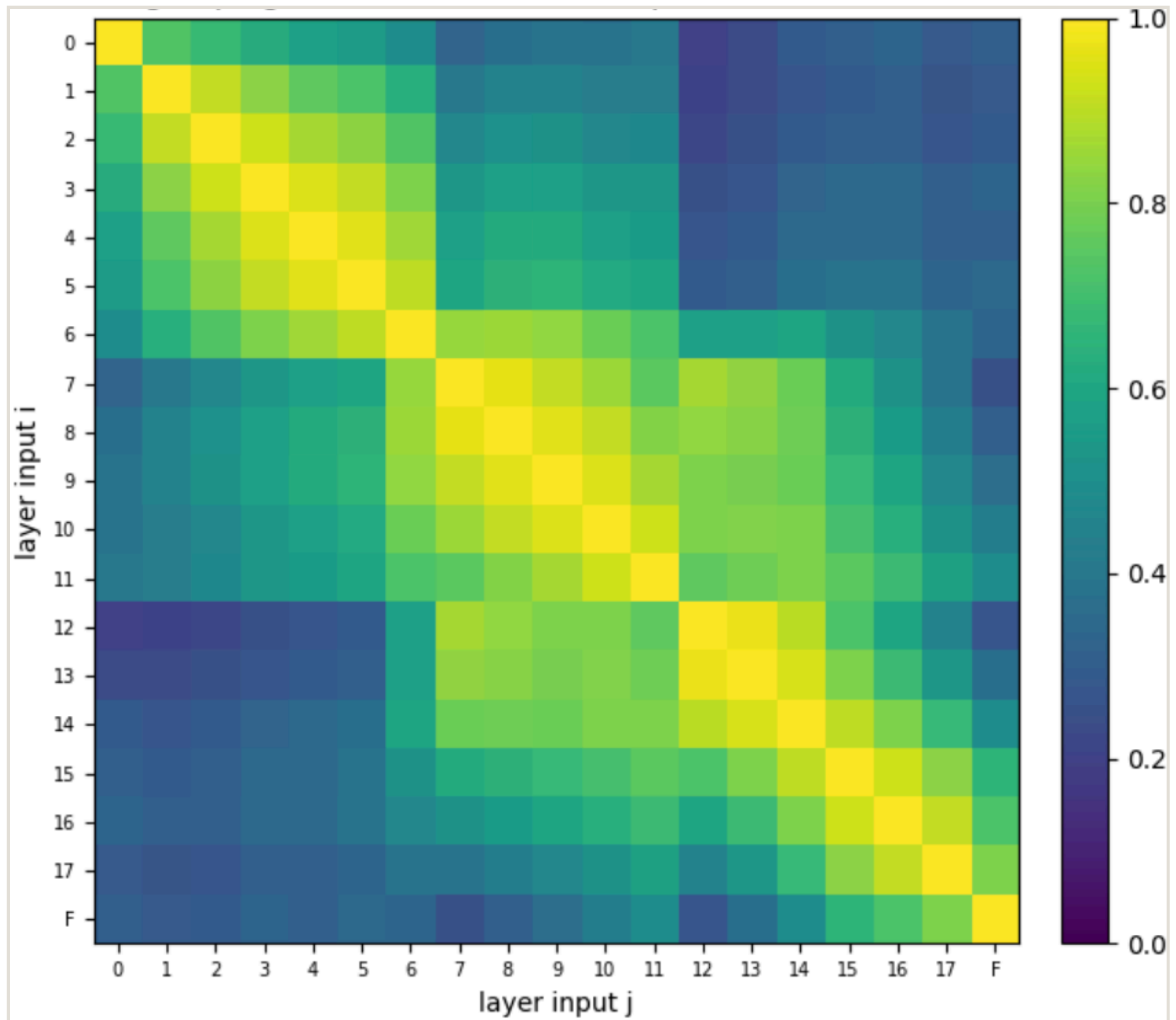


The structure of sharing matters

Variant (R=2, 114M)	PPL@40k
DeepSeek-v3 (no share)	18.83
Random share	18.83
Non-aligned share	18.82
Block share	18.76
Depth embedding only	18.81
Block + Depth (MoRE)	18.71

Block sharing performs the best overall. Unstructured sharing (e.g., Random share, Non-aligned share) collapses to the baseline. In comparison, Block sharing improves noticeably, and **Block + Depth compounds**. The heatmap (right) shows some intuition: **consecutive** layers are similar, hence may effectively share an expert pool.

Why block sharing



CKA similarity between layer inputs (DeepSeek-v3, 114M, 18 layers). Consecutive layers are near-identical. Intuitively, neighbouring layers can share one pool productively.

Lower PPL @ every scale...

Model	114M	369M	1.15B
MoEUT	16.94	14.69	12.78
Dense	16.30	15.34	13.07
DeepSeek-v3	15.98	13.53	11.88
MoRE	15.89	13.40	11.75

and every backbone

Backbone	37M	114M
Qwen3MoE	20.36	15.71
MoRE-Qwen	20.20	15.57
Mixtral	22.45	20.75
MoRE-Mixtral	22.36	20.64

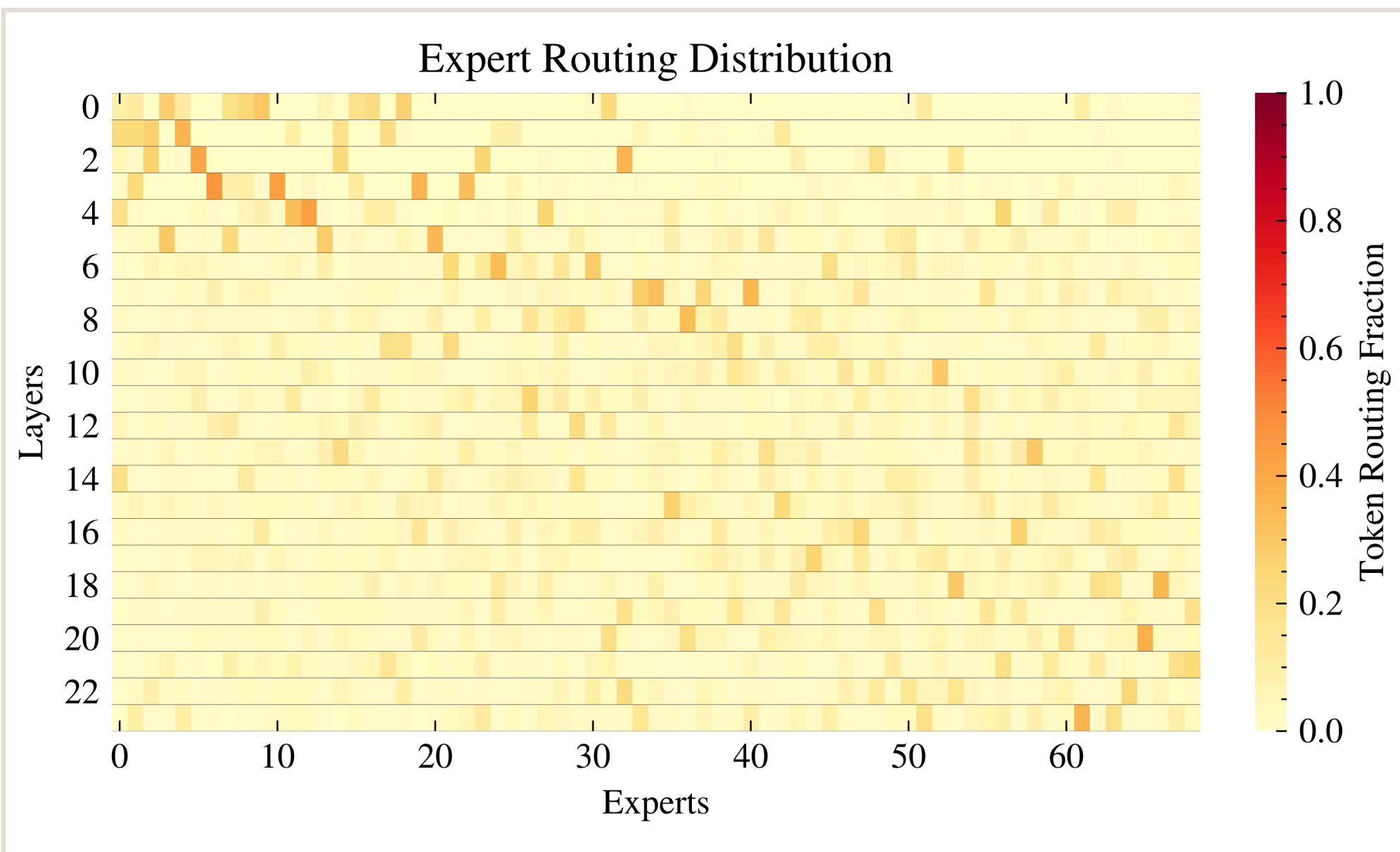
Top performer on downstream tasks

Benchmark	114M				369M				1.15B			
	MoEUT	Dense	DeepSeek	MoRE	MoEUT	Dense	DeepSeek	MoRE	MoEUT	Dense	DeepSeek	MoRE
LAMBADA	16.80	18.51	17.64	19.69	19.46	19.17	22.37	22.60	23.67	23.22	26.87	28.41
HellaSwag	30.51	30.85	31.30	31.59	33.78	31.70	31.77	35.32	37.70	36.52	40.12	40.47
PIQA	60.77	62.45	61.91	61.91	64.63	63.11	62.84	65.12	66.70	65.66	66.75	68.06
ARC-E	35.52	34.80	36.19	36.27	37.04	37.07	38.07	38.76	40.44	40.78	43.77	45.66
OpenBookQA	26.20	25.80	24.60	25.40	25.60	25.40	26.00	27.40	27.20	26.40	28.20	28.60
SciQ	64.80	67.70	66.50	69.70	68.30	67.80	65.90	72.50	70.80	72.90	76.60	77.50
Average	39.10	40.02	39.69	40.76	41.47	40.71	41.16	43.62	44.42	44.25	47.05	48.12

Downstream benchmark accuracy (%) on the OLMo3 evaluation suite, all five-shot. **Bolded** entries within a group are statistically indistinguishable. The bottom row reports the unweighted macro-average across benchmarks.

MoRE genuinely reuses experts

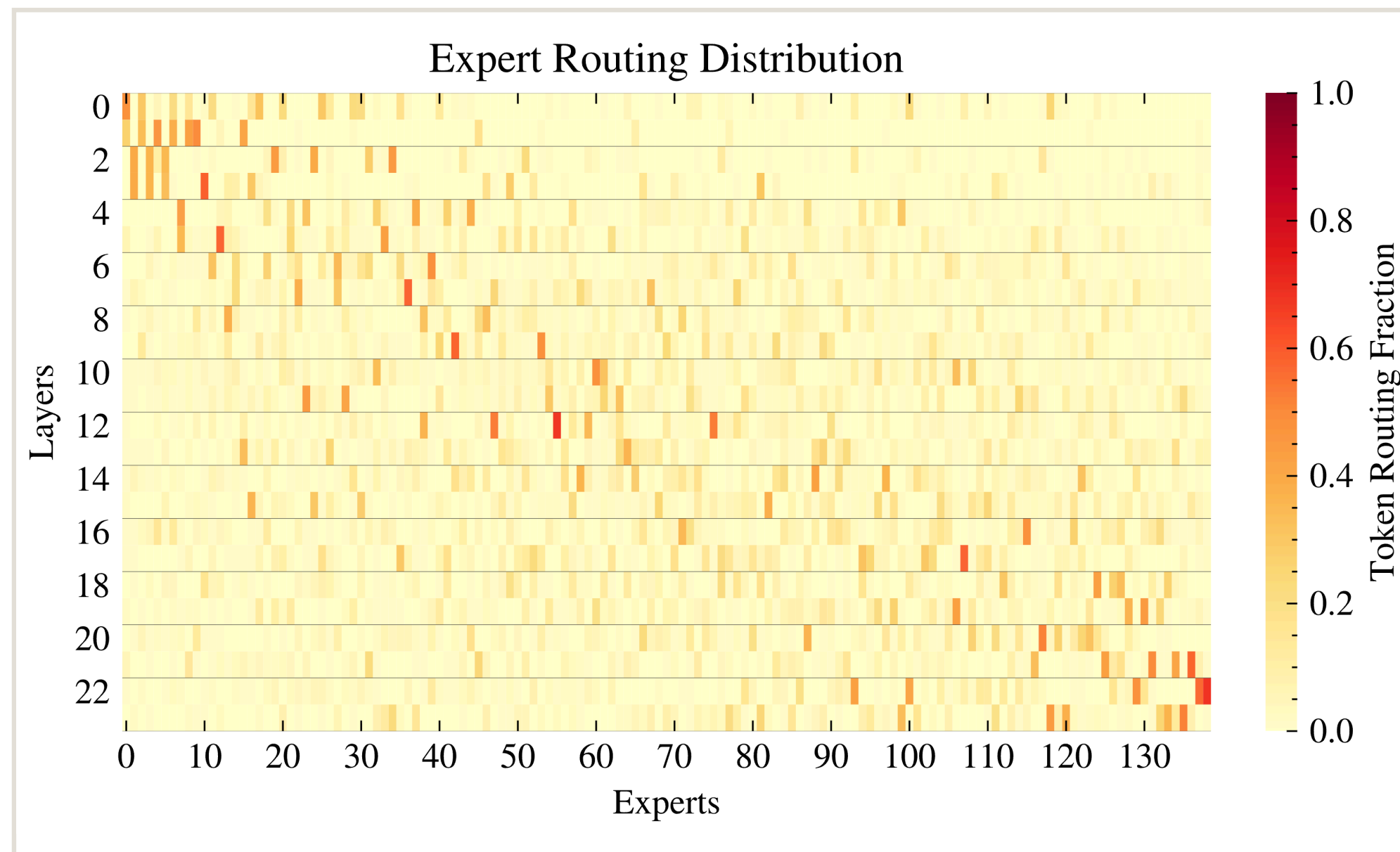
DeepSeek-v3 standard MoE



Standard MoEs only exhibit diagonal depth-specialization band.

8.35% reuse rate, **below** a 10.1% uniform chance baseline

MoRE ours



Additional horizontal streaks across shared layer pairs (0-1, 2-3, ...).

10.3% reuse rate, $\approx 2\times$ the 5.0% uniform chance baseline

For standard MoEs, one routing pattern is impossible: routing to the **same expert across layers**. Yet this expert reuse routing is available to MoRE. The heatmaps (left) compare routing frequency distributions. Compared to standard MoE, MoRE's frequency heatmaps shows **vertical color streaks** across shared pairs, indicating that MoRE exactly routes to the same experts across layers.